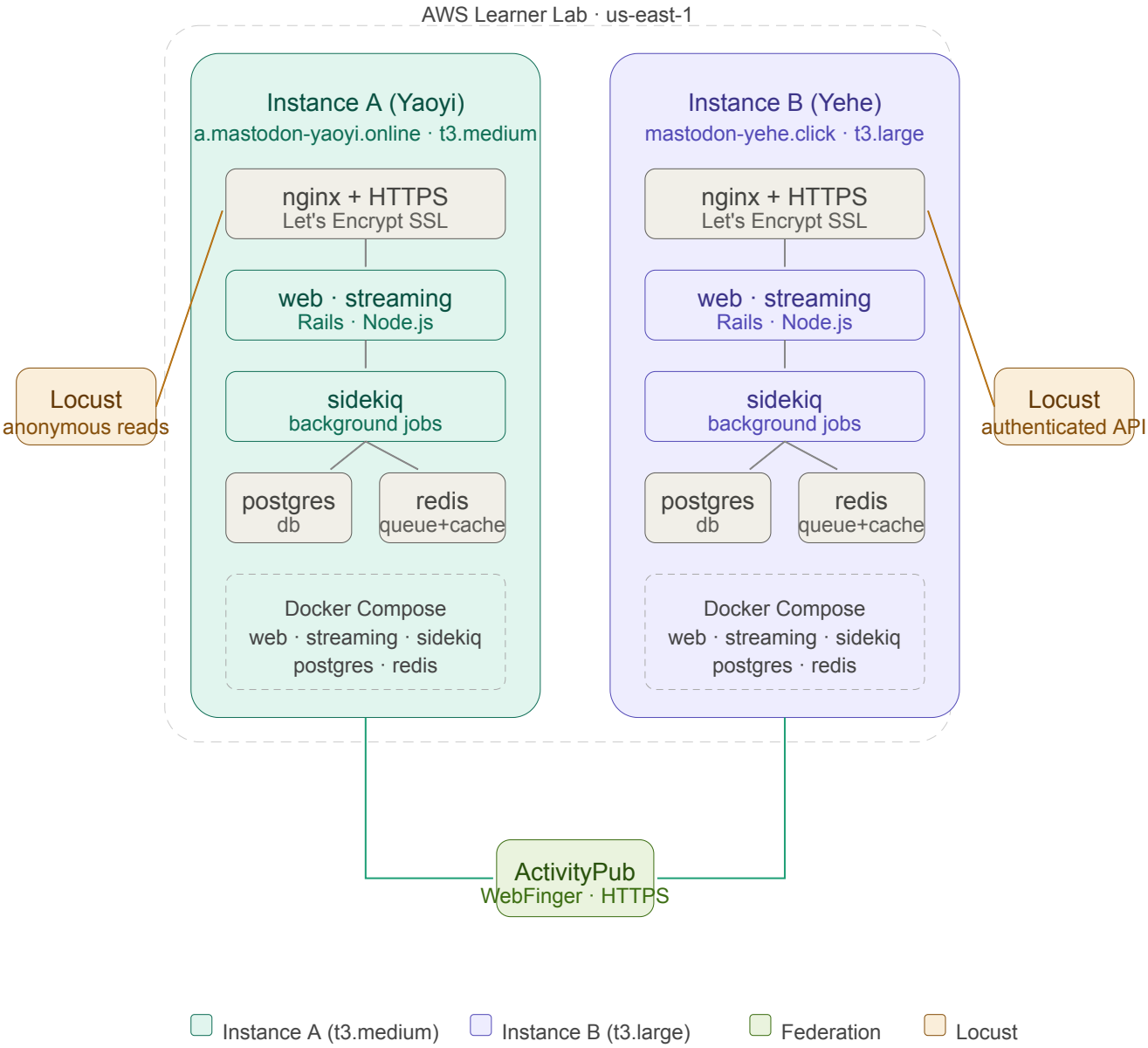
Results and Graphs

Mastodon Scaling Study — CS6650 Spring 2026

Team: Yaoyi Wang & Yehe Yan

Architecture Overview



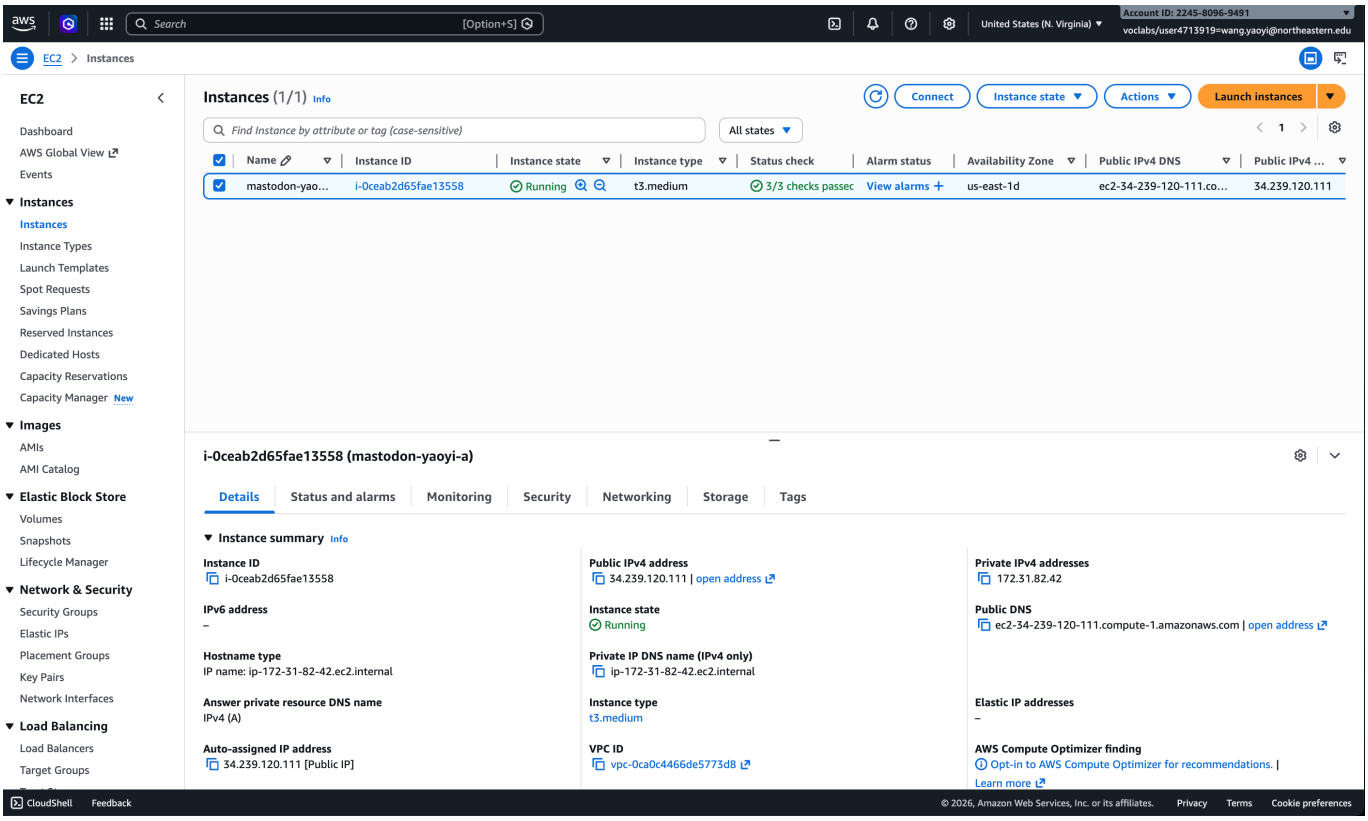
	Instance A (Yaoyi)	Instance B (Yehe)
Domain	`a.mastodon-yaoyi.online`	`mastodon-yehe.click`
EC2	t3.medium (2 vCPU, 4GB RAM)	t3.large (2 vCPU, 8GB RAM)
Workload	Anonymous read-heavy	Authenticated API (read + write)

	Instance A (Yaoyi)	Instance B (Yehe)
Stack	EC2 + Docker Compose + Nginx + HTTPS	EC2 + Docker Compose + Nginx + HTTPS

Experiment 1 — Single Instance Bottleneck (Instance A, Yaoyi)

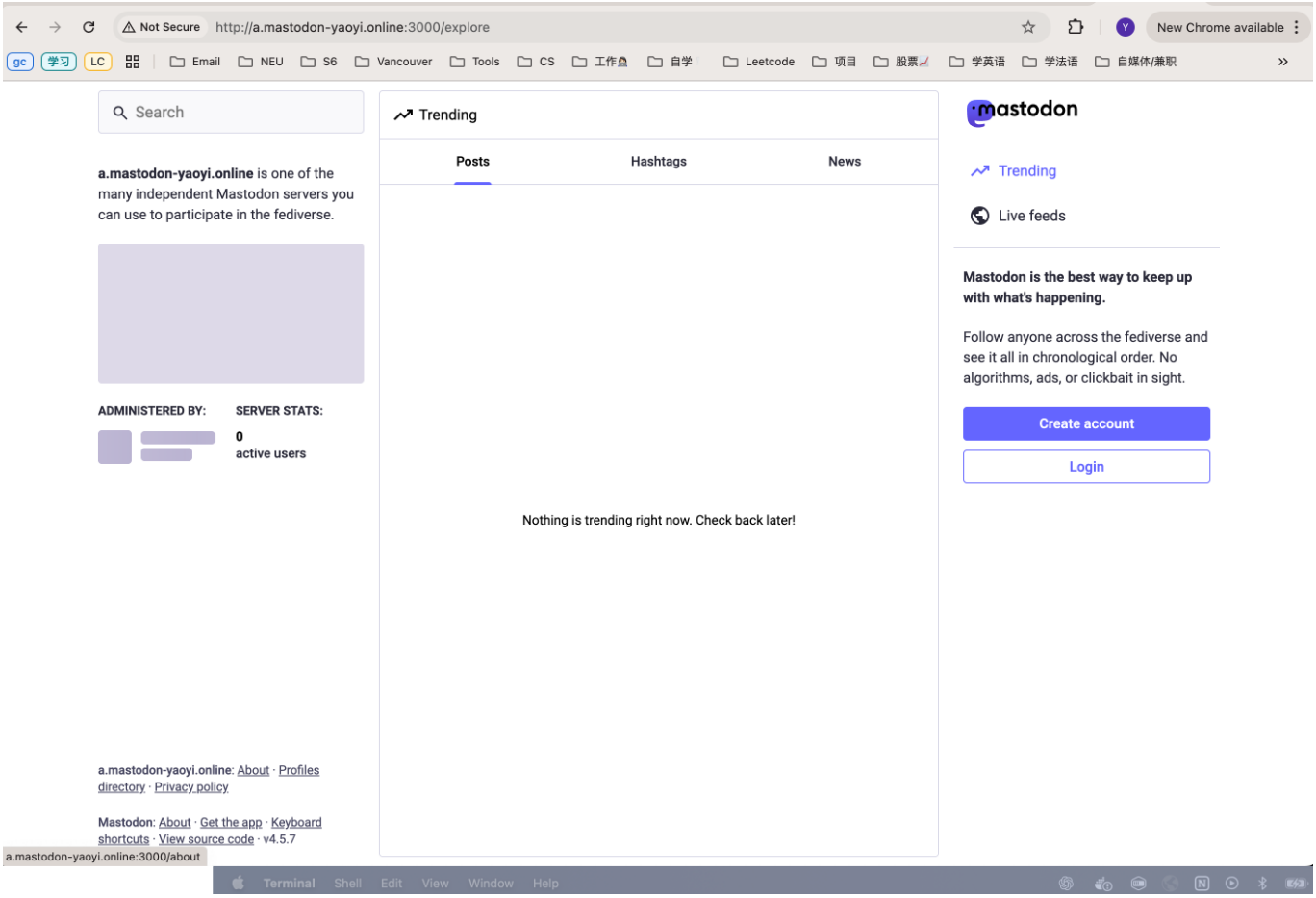
Workload: Anonymous read traffic — `/`, `/about`, `/explore`, `/health` **Tool:** Locust

Deployment — Instance Running



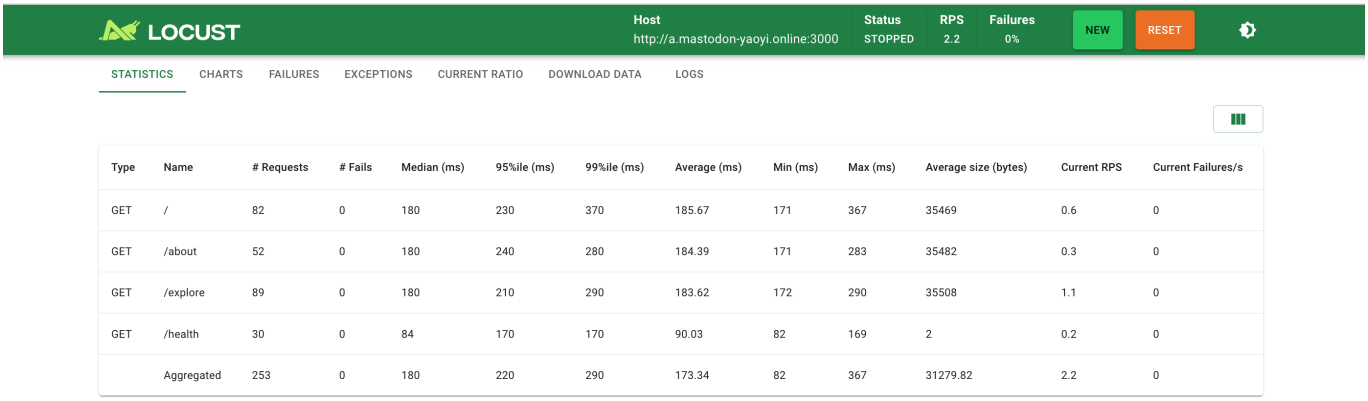
CS 6650 — ubuntu@ip-172-31-82-42: ~/mastodon — ssh -i mastodon-yaoyi-key.pem ubuntu@34.239.120.111...

```
ubuntu@ip-172-31-82-42:~/mastodon$ docker compose up -d db redis
[+] up 28/28
 ✔ Image postgres:14-alpine          Pulled                                                    5.6s
 ✔ Image redis:7-alpine              Pulled                                                    2.8s
 ✔ Network mastodon_internal_network Created                                                   0.0s
 ✔ Container mastodon-redis-1        Started                                                   2.0s
 ✔ Container mastodon-db-1           Started                                                   2.0s
ubuntu@ip-172-31-82-42:~/mastodon$ docker compose ps
NAME                IMAGE                COMMAND                  SERVICE   CREATED         STATUS
           PORTS
mastodon-db-1       postgres:14-alpine   "docker-entrypoint.s…"   db        7 seconds ago   Up 6 seconds (health: s
tarting)   5432/tcp
mastodon-redis-1    redis:7-alpine       "docker-entrypoint.s…"   redis     7 seconds ago   Up 6 seconds (health: s
tarting)   6379/tcp
ubuntu@ip-172-31-82-42:~/mastodon$ docker compose logs --tail=50 db redis
redis-1  | 1:C 22 Mar 2026 20:51:08.063 # WARNING Memory overcommit must be enabled! Without it, a background save
 or replication may fail under low memory condition. Being disabled, it can also cause failures without low memory
 condition, see https://github.com/jemalloc/jemalloc/issues/1328. To fix this issue add 'vm.overcommit_memory = 1'
 to /etc/sysctl.conf and then reboot or run the command 'sysctl vm.overcommit_memory=1' for this to take effect.
redis-1  | 1:C 22 Mar 2026 20:51:08.063 * oO0OoO0OoO0Oo Redis is starting oO0OoO0OoO0Oo
redis-1  | 1:C 22 Mar 2026 20:51:08.063 * Redis version=7.4.8, bits=64, commit=00000000, modified=0, pid=1, just s
tarted
redis-1  | 1:C 22 Mar 2026 20:51:08.064 # Warning: no config file specified, using the default config. In order to
 specify a config file use redis-server /path/to/redis.conf
redis-1  | 1:M 22 Mar 2026 20:51:08.064 * Increased maximum number of open files to 10032 (it was originally set t
o 1024).
redis-1  | 1:M 22 Mar 2026 20:51:08.065 * monotonic clock: POSIX clock_gettime
redis-1  | 1:M 22 Mar 2026 20:51:08.066 * Running mode=standalone, port=6379.
redis-1  | 1:M 22 Mar 2026 20:51:08.066 * Server initialized
redis-1  | 1:M 22 Mar 2026 20:51:08.067 * Ready to accept connections tcp
db-1     | Data page checksums are disabled.
db-1     |
db-1     | fixing permissions on existing directory /var/lib/postgresql/data ... ok
db-1     | creating subdirectories ... ok
db-1     | selecting dynamic shared memory implementation ... posix
db-1     | selecting default max_connections ... 100
db-1     | selecting default shared_buffers ... 128MB
db-1     | selecting default time zone ... UTC
db-1     | creating configuration files ... ok
db-1     | running bootstrap script ... ok
db-1     | sh: locale: not found
db-1     | 2026-03-22 20:51:08.626 UTC [35] WARNING:  no usable system locales were found
db-1     | performing post-bootstrap initialization ... ok
db-1     | initdb: warning: enabling "trust" authentication for local connections
db-1     | You can change this by editing pg_hba.conf or using the option -A, or
db-1     | --auth-local and --auth-host, the next time you run initdb.
db-1     | syncing data to disk ... ok
db-1     |
db-1     |
db-1     | Success. You can now start the database server using:
db-1     |
db-1     |     pg_ctl -D /var/lib/postgresql/data -l logfile start
db-1     |
db-1     | waiting for server to start....2026-03-22 20:51:09.373 UTC [41] LOG:  starting PostgreSQL 14.22 on x86_
64-pc-linux-musl, compiled by gcc (Alpine 15.2.0) 15.2.0, 64-bit
db-1     | 2026-03-22 20:51:09.376 UTC [41] LOG:  listening on Unix socket "/var/run/postgresql/.s.PGSQL.5432"
db-1     | 2026-03-22 20:51:09.387 UTC [42] LOG:  database system was shut down at 2026-03-22 20:51:09 UTC
db-1     | 2026-03-22 20:51:09.395 UTC [41] LOG:  database system is ready to accept connections
db-1     |  done
db-1     | server started
db-1     | CREATE DATABASE
```



Load Test Results

5 Users



Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/	82	0	180	230	370	185.67	171	367	35469	0.6	0
GET	/about	52	0	180	240	280	184.39	171	283	35482	0.3	0
GET	/explore	89	0	180	210	290	183.62	172	290	35508	1.1	0
GET	/health	30	0	84	170	170	90.03	82	169	2	0.2	0
	Aggregated	253	0	180	220	290	173.34	82	367	31279.82	2.2	0

20 Users

LOCUST

Host: http://a.mastodon-yaoyi.online:3000 | Status: STOPPED | RPS: 9.4 | Failures: 0%

NEW RESET

STATISTICS CHARTS FAILURES EXCEPTIONS CURRENT RATIO DOWNLOAD DATA LOGS

Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/	347	0	180	210	300	183.16	170	501	35469	3	0
GET	/about	228	0	180	190	270	179.89	169	293	35482	1.9	0
GET	/explore	344	0	180	210	290	183.78	170	424	35508	3.3	0
GET	/health	133	0	83	89	170	85.64	80	172	2	1.2	0
	Aggregated	1052	0	180	200	290	170.32	80	501	31000.62	9.4	0

50 Users

LOCUST

Host: http://a.mastodon-yaoyi.online:3000 | Status: STOPPED | RPS: 22.8 | Failures: 0%

NEW RESET

STATISTICS CHARTS FAILURES EXCEPTIONS CURRENT RATIO DOWNLOAD DATA LOGS

Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/	757	0	180	230	570	194.13	169	1054	35469	8.7	0
GET	/about	510	0	180	280	840	200.78	170	1143	35482	4	0
GET	/explore	801	0	180	270	830	200.54	169	1260	35508	7.7	0
GET	/health	283	0	84	97	200	91.64	80	794	2	2.4	0
	Aggregated	2351	0	180	240	670	185.42	80	1260	31215.79	22.8	0

100 Users

LOCUST

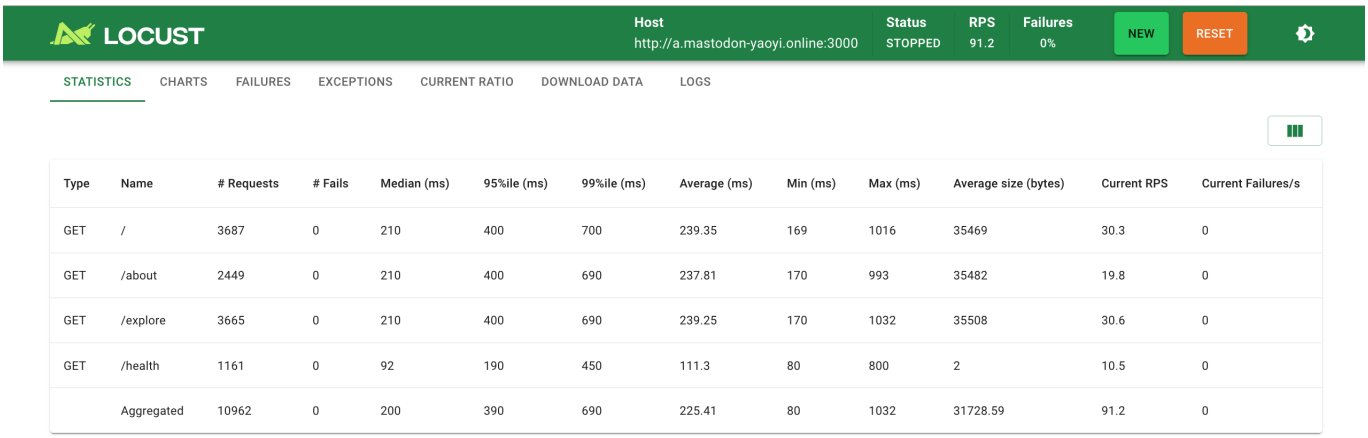
Host: http://a.mastodon-yaoyi.online:3000 | Status: STOPPED | RPS: 46.6 | Failures: 0%

NEW RESET

STATISTICS CHARTS FAILURES EXCEPTIONS CURRENT RATIO DOWNLOAD DATA LOGS

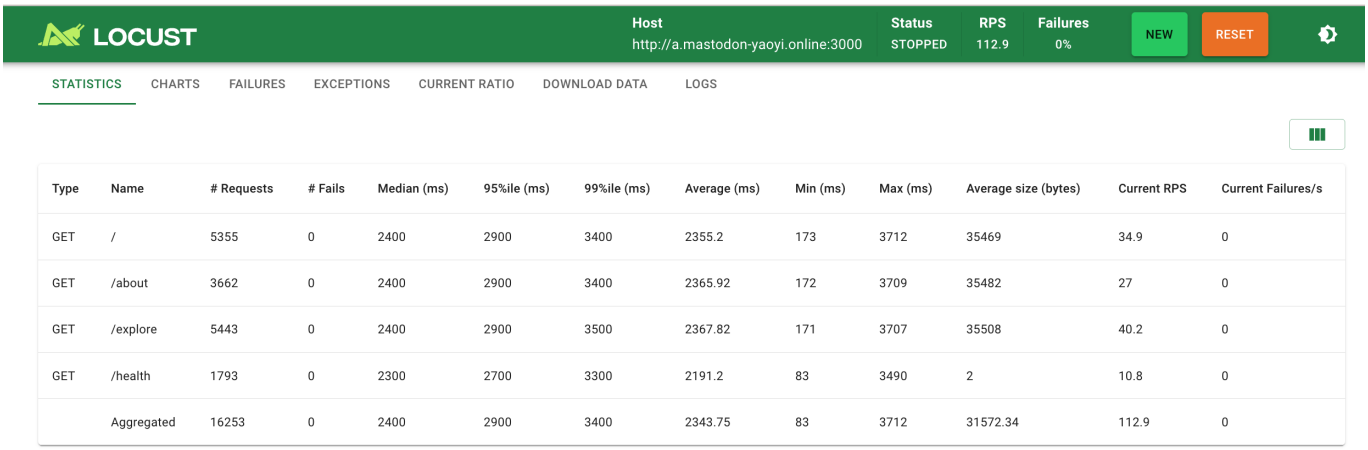
Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/	2134	0	180	230	370	189.96	169	530	35469	15.6	0
GET	/about	1416	0	180	230	380	189.73	170	522	35482	10.5	0
GET	/explore	2169	0	180	230	400	190.34	170	499	35508	16	0
GET	/health	706	0	84	100	220	88.59	79	253	2	4.5	0
	Aggregated	6425	0	180	220	370	178.9	79	530	31587.8	46.6	0

200 Users



Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/	3687	0	210	400	700	239.35	169	1016	35469	30.3	0
GET	/about	2449	0	210	400	690	237.81	170	993	35482	19.8	0
GET	/explore	3665	0	210	400	690	239.25	170	1032	35508	30.6	0
GET	/health	1161	0	92	190	450	111.3	80	800	2	10.5	0
	Aggregated	10962	0	200	390	690	225.41	80	1032	31728.59	91.2	0

500 Users



Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/	5355	0	2400	2900	3400	2355.2	173	3712	35469	34.9	0
GET	/about	3662	0	2400	2900	3400	2365.92	172	3709	35482	27	0
GET	/explore	5443	0	2400	2900	3500	2367.82	171	3707	35508	40.2	0
GET	/health	1793	0	2300	2700	3300	2191.2	83	3490	2	10.8	0
	Aggregated	16253	0	2400	2900	3400	2343.75	83	3712	31572.34	112.9	0

Summary Table

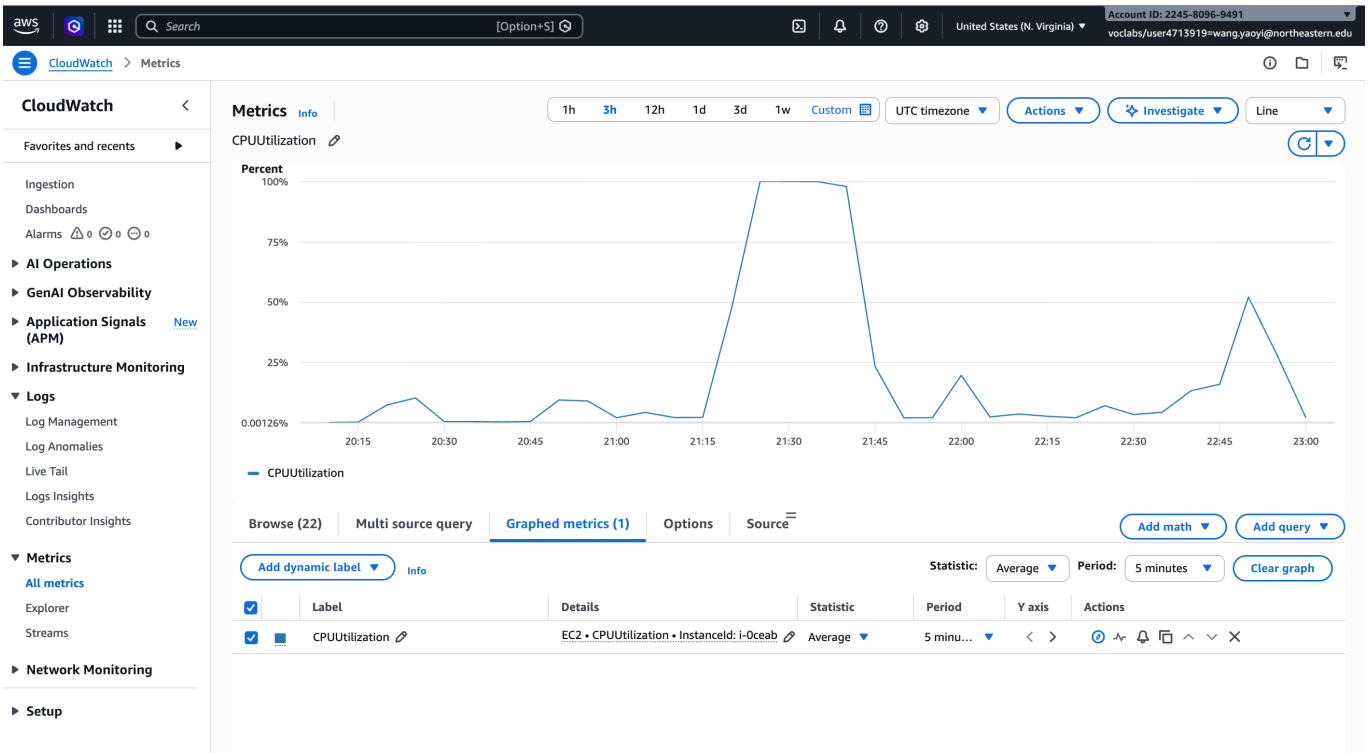
Load Level	Avg Latency (ms)	P95 (ms)	P99 (ms)	RPS	Failures
5 users	173	220	290	2.2	0 (0%)
20 users	170	200	290	9.4	0 (0%)
50 users	185	240	670	22.8	0 (0%)
100 users	179	220	370	46.6	0 (0%)
200 users	225	390	690	91.2	0 (0%)
500 users	2344	2900	3400	112.9	0 (0%)

Key finding: Zero failures at all load levels. Throughput grew steadily up to 200 users, then plateaued at 500 users while latency increased sharply (225ms → 2344ms avg), indicating the instance approached saturation around 200–500 users. Web container showed the highest CPU usage under load.

Docker Stats Under Load

```
CS 6650 — ubuntu@ip-172-31-82-42: ~/mastodon — ssh -i mastodon-yaoyi-key.pem ubuntu@34.239.120.111...
ubuntu@ip-172-31-82-42:~/mastodon$ date && docker stats --no-stream
Sun Mar 22 22:36:04 UTC 2026
CONTAINER ID   NAME                   CPU %     MEM USAGE / LIMIT     MEM %     NET I/O           BLOCK I/O
  PIDS
a3593b4e239c   mastodon-sidekiq-1     0.17%     343.8MiB / 3.744GiB   8.97%     1.5MB / 2.45MB    958kB / 15.1MB
  21
5562e255a112   mastodon-web-1         2.63%     695.1MiB / 3.744GiB   18.13%    3.35MB / 19.4MB   57.1MB / 15.1MB
  33
4abe565f3953   mastodon-streaming-1   0.55%     53.34MiB / 3.744GiB   1.39%     16.2kB / 2.82kB   2.9MB / 0B
  11
19a0974accac   mastodon-db-1          3.91%     61.05MiB / 3.744GiB   1.59%     1.04MB / 2.34MB   4.37MB / 6.84MB
  17
588bddeb3492   mastodon-redis-1       4.37%     4.957MiB / 3.744GiB   0.13%     3.94MB / 2.08MB   778kB / 741kB
  6
ubuntu@ip-172-31-82-42:~/mastodon$ date && docker stats --no-stream
Sun Mar 22 22:54:09 UTC 2026
CONTAINER ID   NAME                   CPU %     MEM USAGE / LIMIT     MEM %     NET I/O           BLOCK I/O
  PIDS
a3593b4e239c   mastodon-sidekiq-1     0.10%     343MiB / 3.744GiB     8.95%     2.12MB / 3.7MB    958kB / 15.1MB
  21
5562e255a112   mastodon-web-1         168.23%   816.8MiB / 3.744GiB   21.30%    93MB / 1.06GB     57.2MB / 15.1MB
  33
4abe565f3953   mastodon-streaming-1   0.43%     53.62MiB / 3.744GiB   1.40%     17.1kB / 3.68kB   2.9MB / 0B
  11
19a0974accac   mastodon-db-1          8.42%     64.47MiB / 3.744GiB   1.68%     20.2MB / 20.6MB   4.37MB / 9.97MB
  19
588bddeb3492   mastodon-redis-1       10.59%    5.363MiB / 3.744GiB   0.14%     95.5MB / 58.1MB   778kB / 1.46MB
  6
ubuntu@ip-172-31-82-42:~/mastodon$
```

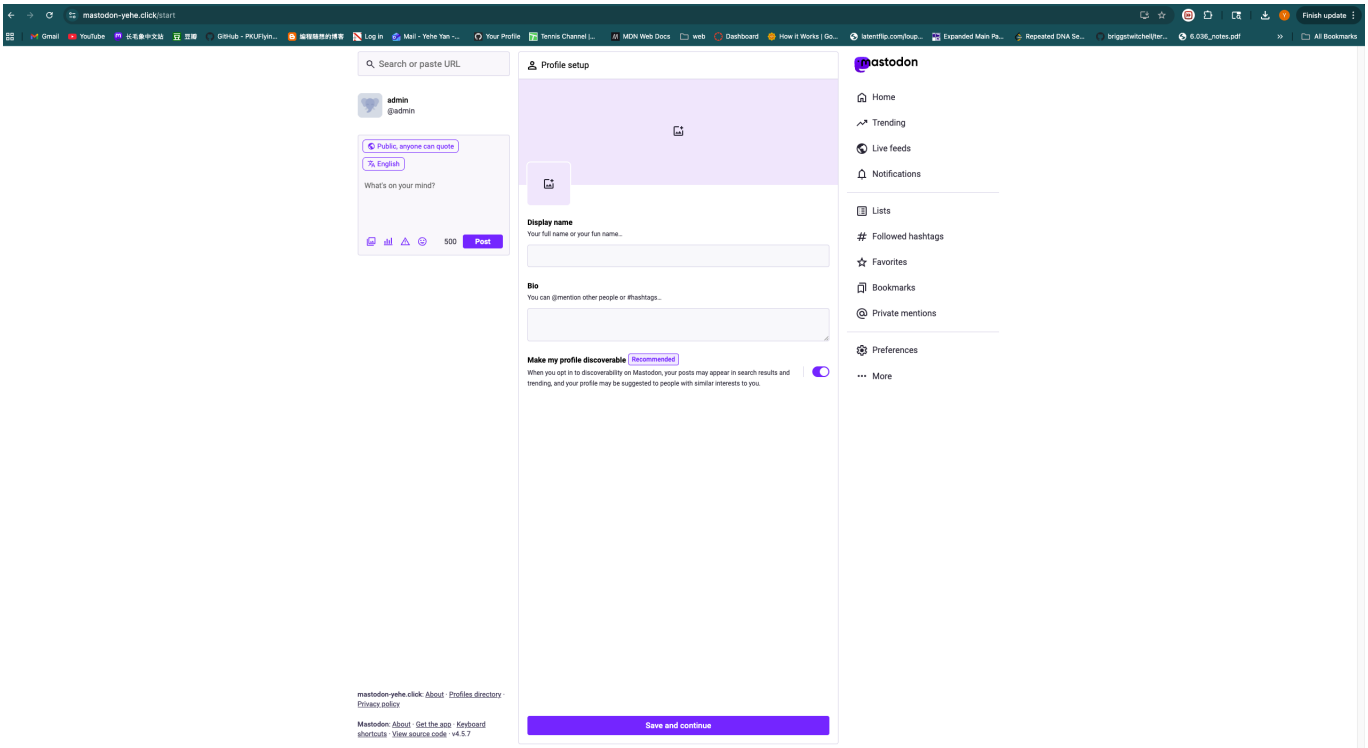
CloudWatch CPU

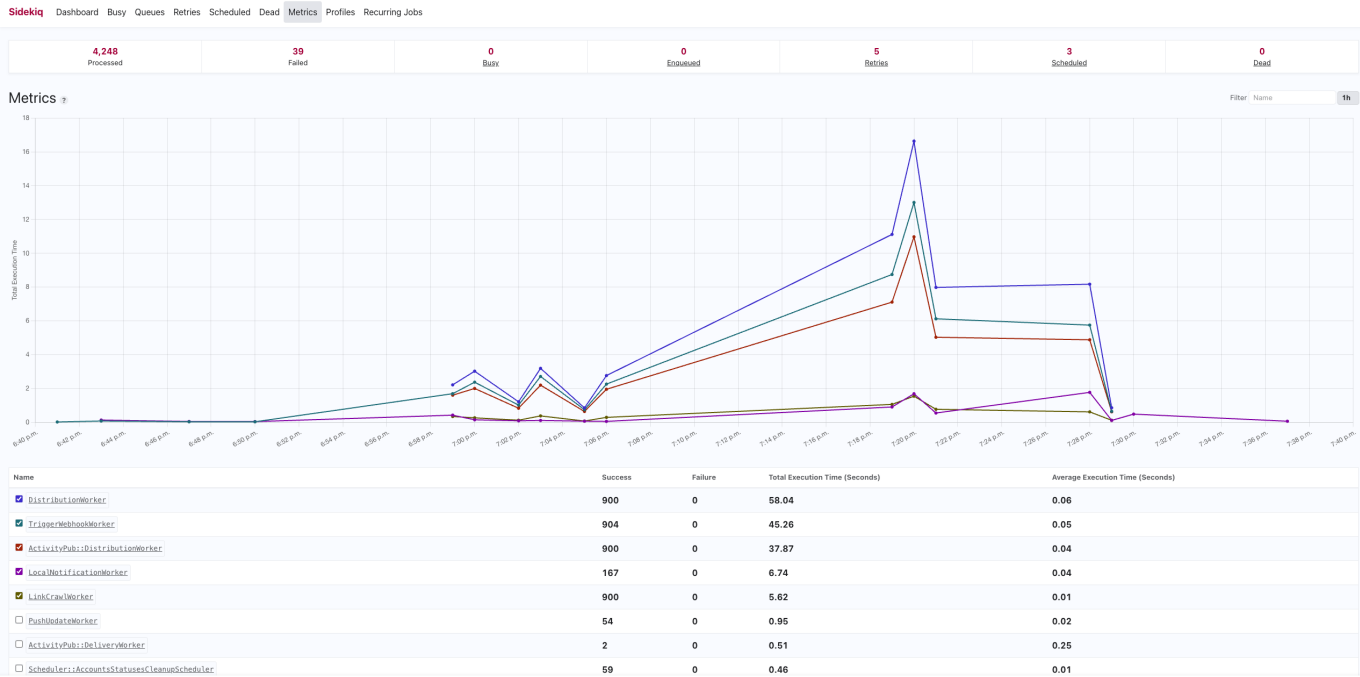


Experiment 2 — Authenticated API Load Test + Rate Limiter Behavior (Instance B, Yehe)

Workload: Authenticated API — POST status (3x), GET home timeline (5x), GET public timeline (3x), GET notifications (2x), favourite (1x), search (1x) **Tool:** Locust

Instance Running





Load Test Results

Smoke Test — 5 Users, 60s

Endpoint	Requests	Failures	p50	p95	p99	RPS
POST /api/v1/statuses	77	0 (0%)	95ms	150ms	310ms	1.30
GET /timelines/home	67	0 (0%)	180ms	280ms	410ms	1.14
GET /timelines/public	22	0 (0%)	170ms	210ms	370ms	0.37

Endpoint	Requests	Failures	p50	p95	p99	RPS
GET /notifications	22	0 (0%)	110ms	190ms	240ms	0.37
GET /api/v2/search	11	0 (0%)	62ms	180ms	180ms	0.19
Aggregated	**213**	**0 (0%)**	**130ms**	**230ms**	**310ms**	**3.61**

Baseline — 20 Users, 300s

Endpoint	Requests	Failures	Failure %	p50	p95	p99	RPS
POST /api/v1/statuses	1,383	834	60.3%	82ms	400ms	670ms	4.62
GET /timelines/home	1,153	668	57.9%	100ms	470ms	790ms	3.85
GET /timelines/public	591	280	47.4%	170ms	430ms	700ms	1.97
GET /notifications	404	215	53.2%	110ms	570ms	970ms	1.35
GET /api/v2/search	184	103	56.0%	61ms	200ms	310ms	0.61
Aggregated	**4,041**	**2,226**	**55.1%**	**94ms**	**440ms**	**740ms**	**13.49**

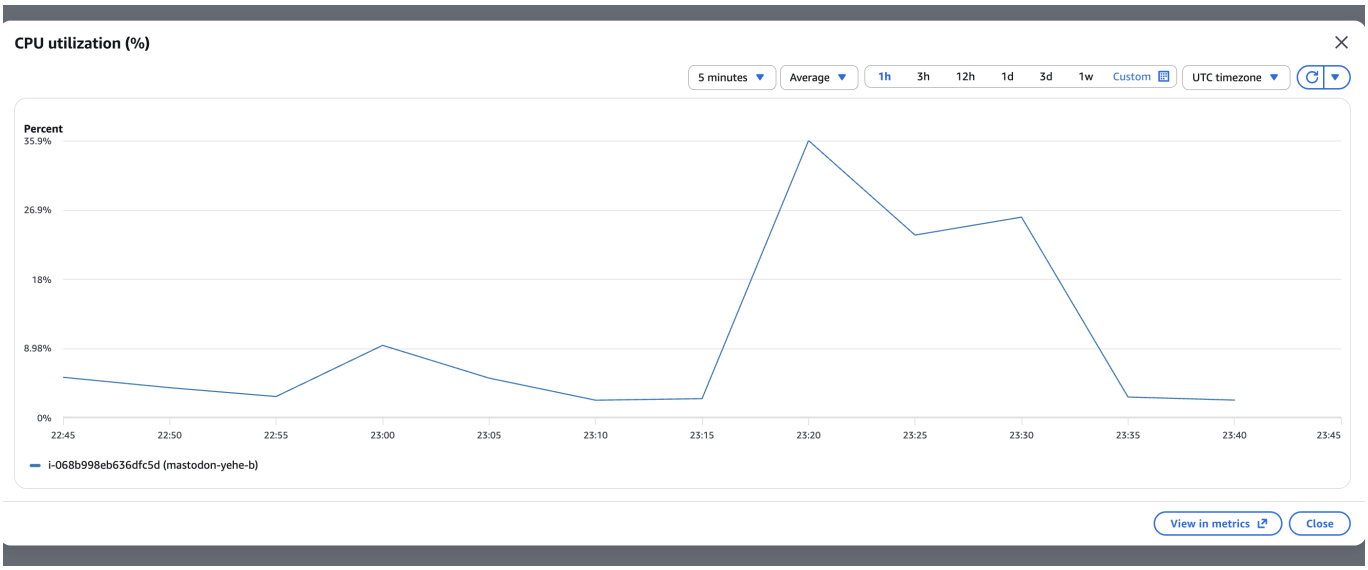
All failures: HTTP 429 Too Many Requests.

Stress Test — 50 Users, 300s

Endpoint	Requests	Failures	Failure %	p50	p95
POST /api/v1/statuses	3,129	3,029	96.8%	92ms	1,300ms
GET /timelines/home	2,669	2,011	75.3%	91ms	1,700ms
GET /timelines/public	1,276	819	64.2%	100ms	1,500ms
GET /notifications	853	663	77.7%	89ms	1,900ms
GET /api/v2/search	474	344	72.6%	96ms	1,200ms
Aggregated	**8,990**	**7,212**	**80.2%**	**94ms**	**1,500ms**

All failures: HTTP 429 Too Many Requests.

EC2 CPU During Load Tests



Test	EC2 CPU Peak	Web Container CPU	DB CPU	Sidekiq
5 users (smoke)	~5%	normal	normal	processing
20 users (baseline)	35.9%	elevated	11%	processing
50 users (stress)	~26%	177% (1 core)	11%	idle

Load Test Progression Summary

Load Level	Error Rate	p50	p95	RPS	Bottleneck
5 users (smoke)	0%	130ms	230ms	3.6	None
20 users (baseline)	55.1%	94ms	440ms	13.5	Rate limiter begins
50 users (stress)	80.2%	94ms	1,500ms	30.0	Rate limiter + web CPU 177%

Key findings:

- Rate limiter (`rack-attack`) activates before hardware saturation — 429s appear at 20 users
- Web container hit 177% CPU (one full core) at 50 users while DB stayed at 11%
- Sidekiq starved: no successful POST = no jobs enqueued = Sidekiq idle
- p50 latency improved at 50 users (130ms → 94ms) because instant 429 rejections pulled the median down
- **Next step:** disable rate limiting (`RACK_ATTACK_ENABLED=false`) to find true hardware saturation point

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Experiment 3 — Federation Validation (Instance A ↔ Instance B)

Federation was validated after Instance A was upgraded to HTTPS + Nginx (Let's Encrypt).

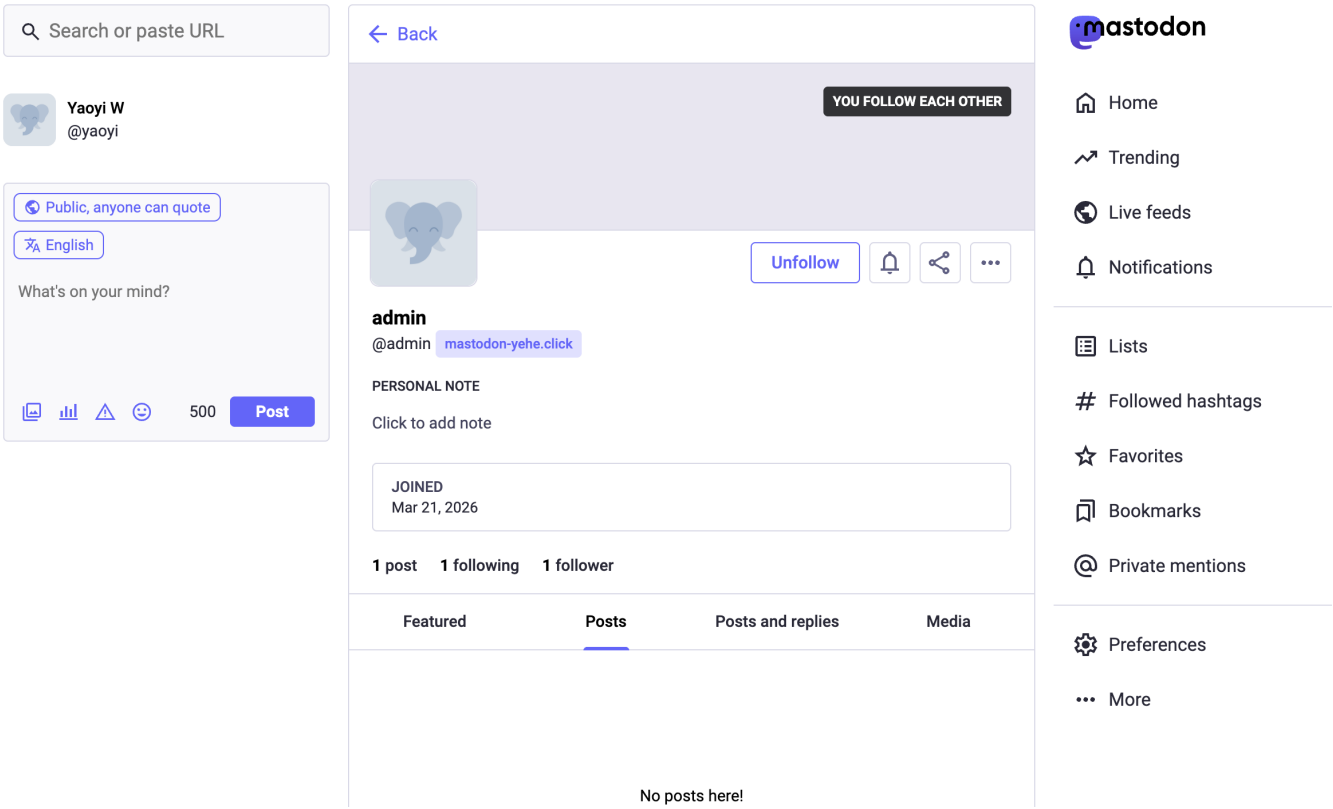
Federation Checks

Check	Result

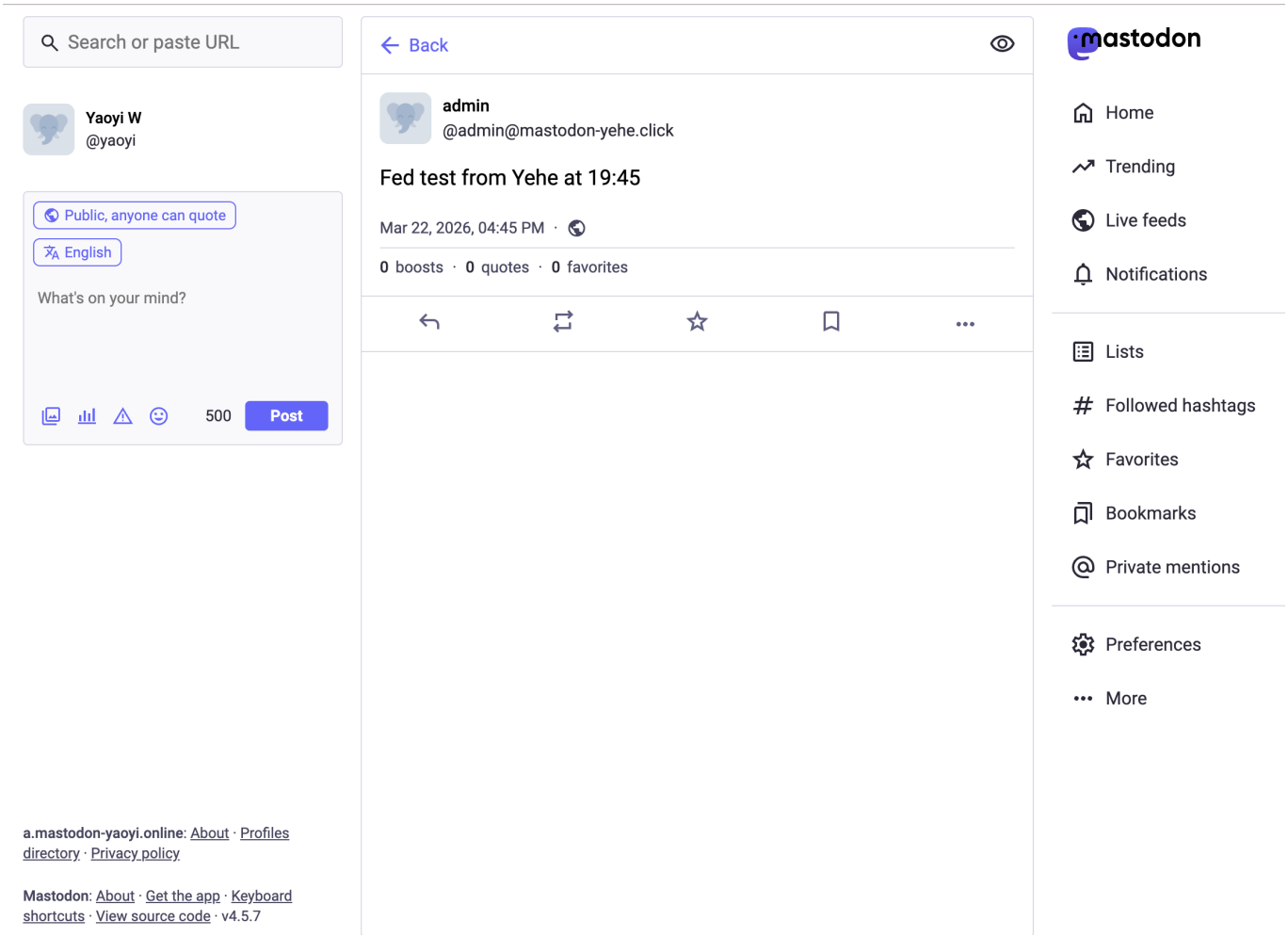
Check	Result
Remote account discovery (`@admin@mastodon-yehe.click` from Instance A)	✅ Success
Mutual follow between instances	✅ Success
Remote public post visible on home timeline	✅ Success
Remote like / favorite notification received	✅ Success

Screenshots

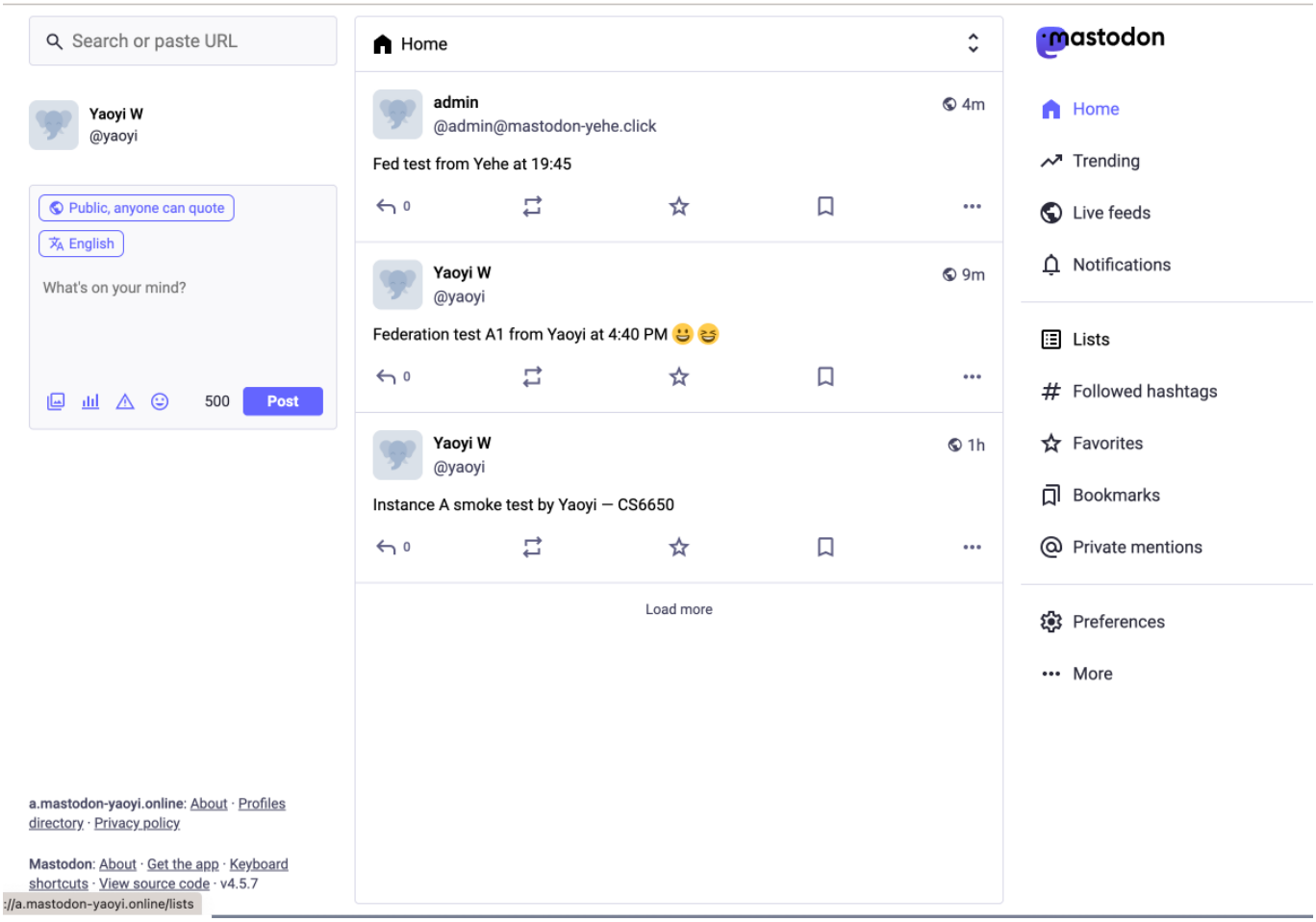
Mutual Follow



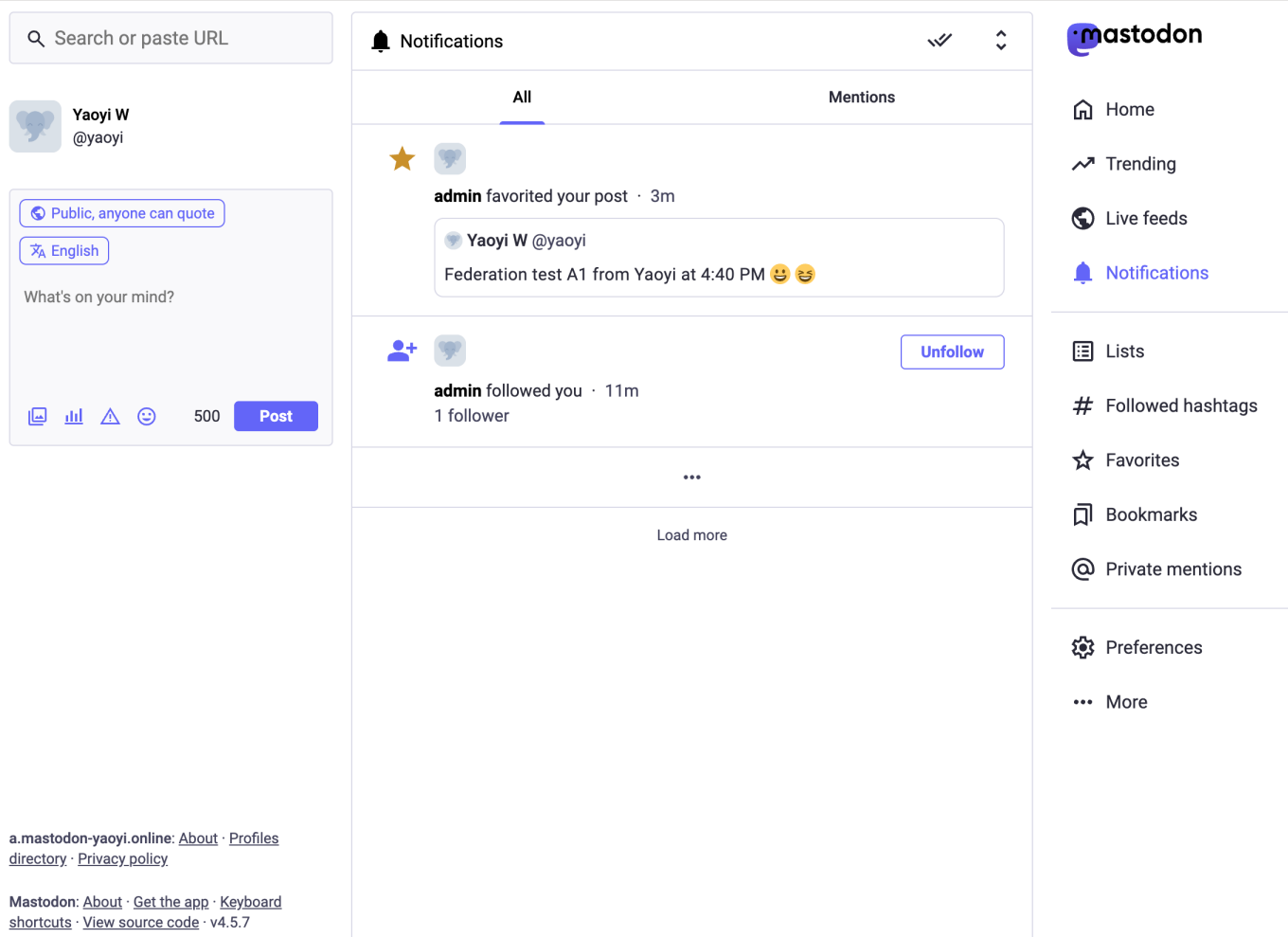
Remote Profile Visible



Remote Post Visible



Remote Like Notification



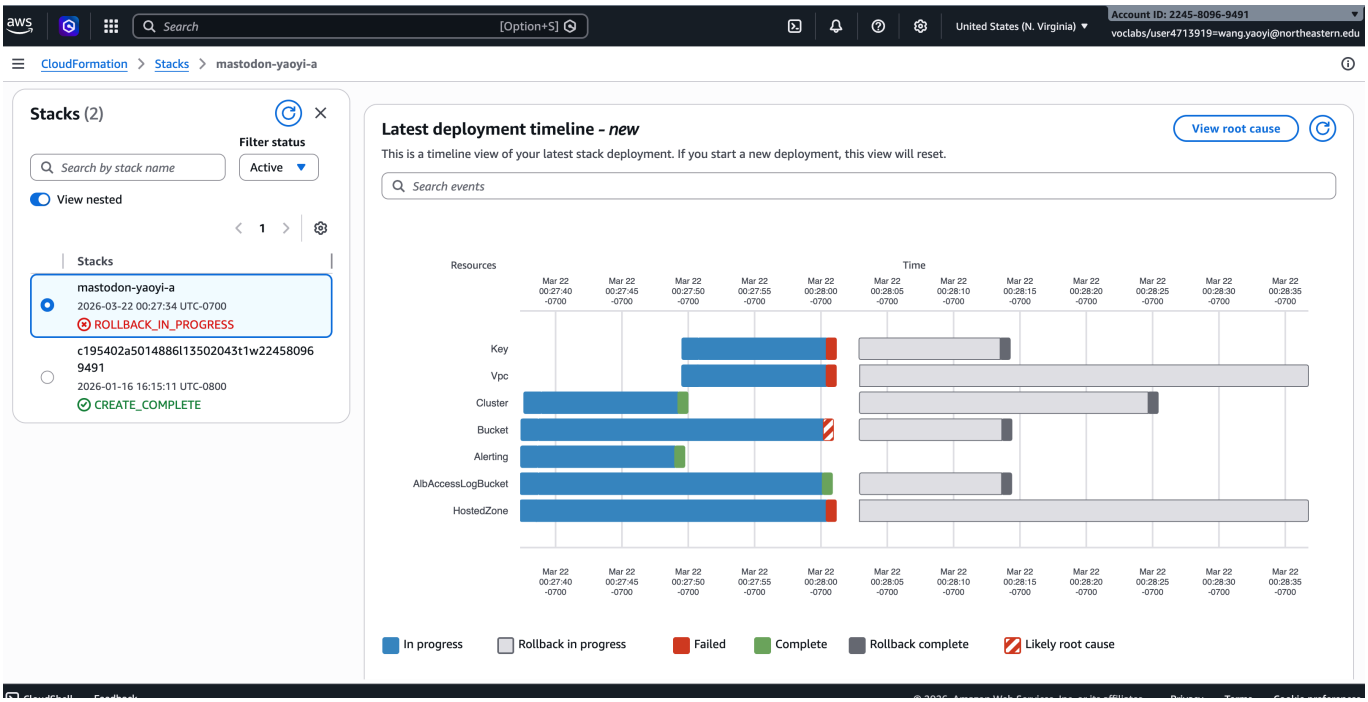
Key finding: Federation required HTTPS + Nginx — direct HTTP on port 3000 was insufficient for WebFinger / ActivityPub discoverability. Once both instances were production-configured, all four federation checks passed. Quantitative propagation latency measurement is a next step.

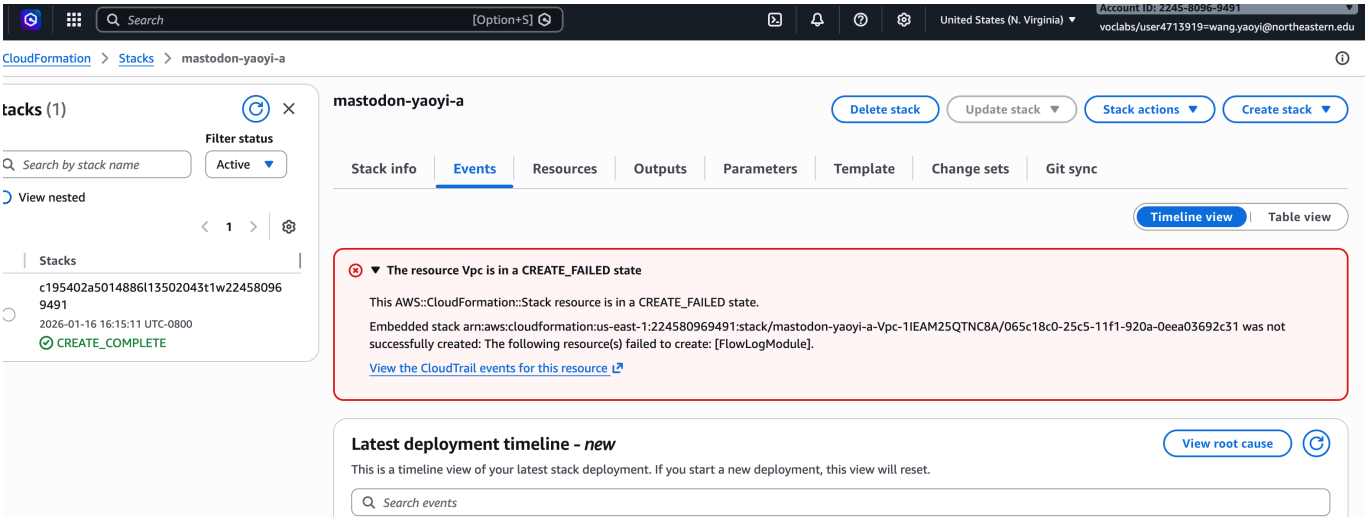
Deployment Pivot — CloudFormation Failure Evidence

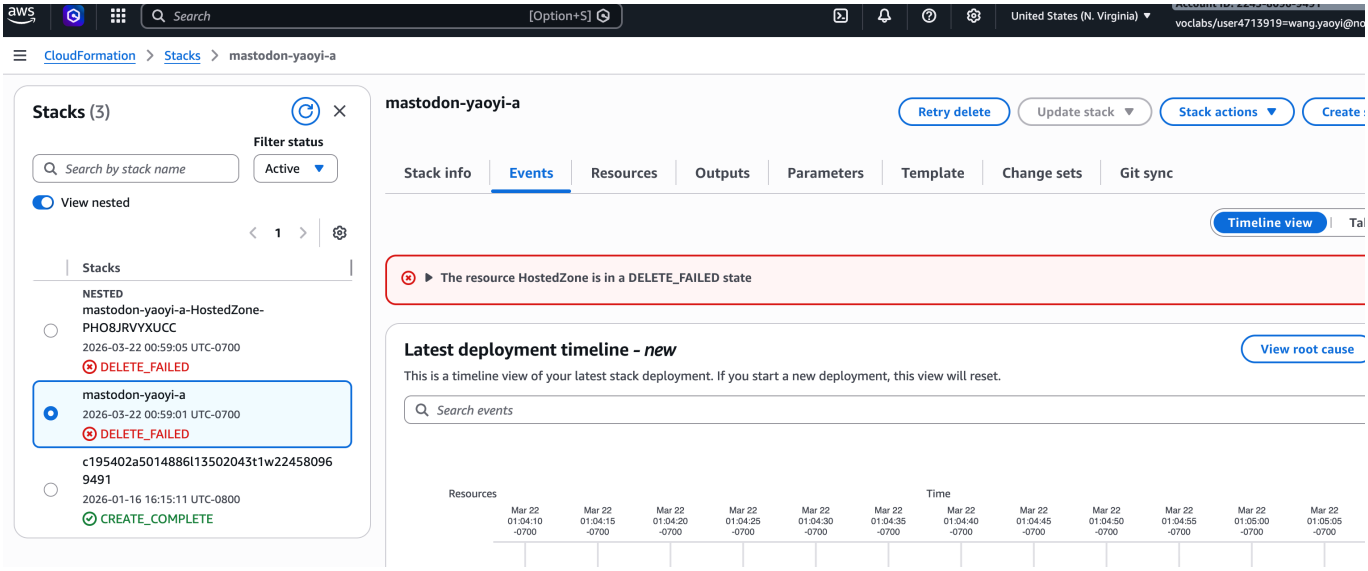
The original deployment path (CloudFormation + ECS Fargate) was abandoned after 5 failed attempts due to Learner Lab IAM restrictions.

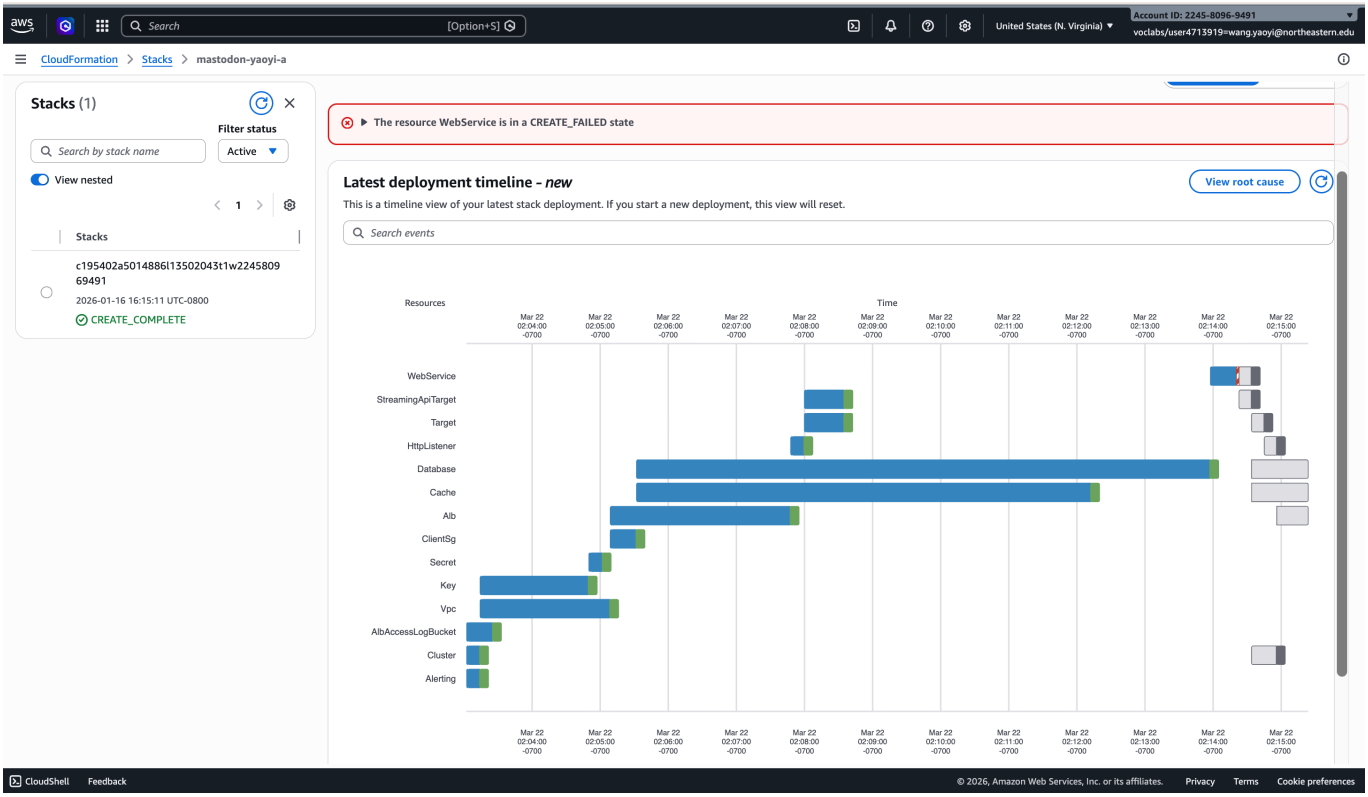
Template Version	Failed Resource	Root Cause
v1	`EmailIdentity`, `LambdaRole`	SES + IAM role creation blocked
v2	`BucketPolicyPublic`	S3 Block Public Access
v3/v4	`FlowLogModule`	VPC flow logs require IAM role creation
v5	`TaskRole` / `TaskExecutionRole`	ECS task IAM roles fundamentally blocked

Stack Failure Screenshots


Search [Option+S]
United States (N. Virginia)
Account ID: 2245-8096-9491
voclabs/user4713919=wang.yaoyi@northeastern.edu
CloudFormation > Stacks > mastodon-yaoyi-a
Stacks (2)
Filter status
Search by stack name
Active
View nested
Stacks
mastodon-yaoyi-a
2026-03-22 00:27:34 UTC-0700
ROLLBACK_IN_PROGRESS
c195402a5014886l13502043t1w22458096 9491
2026-01-16 16:15:11 UTC-0800
CREATE_COMPLETE
Latest deployment timeline - new
View root cause
This is a timeline view of your latest stack deployment. If you start a new deployment, this view will reset.
Search events
Resources
Time
Key
Vpc
Cluster
Bucket
Alerting
AlbAccessLogBucket
HostedZone
In progress
Rollback in progress
Failed
Complete
Rollback complete
Likely root cause


Search [Option+S]
United States (N. Virginia)
Account ID: 2245-8096-9491
voclabs/user4713919=wang.yaoyi@northeastern.edu
CloudFormation > Stacks > mastodon-yaoyi-a
tacks (1)
Filter status
Search by stack name
Active
View nested
Stacks
c195402a5014886l13502043t1w22458096 9491
2026-01-16 16:15:11 UTC-0800
CREATE_COMPLETE
mastodon-yaoyi-a
Delete stack
Update stack
Stack actions
Create stack
Stack info
Events
Resources
Outputs
Parameters
Template
Change sets
Git sync
Timeline view
Table view
The resource Vpc is in a CREATE_FAILED state
This AWS::CloudFormation::Stack resource is in a CREATE_FAILED state.
Embedded stack arn:aws:cloudformation:us-east-1:224580969491:stack/mastodon-yaoyi-a-Vpc-1IEAM25QTNC8A/065c18c0-25c5-11f1-920a-0eea03692c31 was not successfully created: The following resource(s) failed to create: [FlowLogModule].
View the CloudTrail events for this resource
Latest deployment timeline - new
View root cause
This is a timeline view of your latest stack deployment. If you start a new deployment, this view will reset.
Search events


Search [Option+S]
United States (N. Virginia)
CloudFormation > Stacks > mastodon-yaoyi-a
Stacks (3)
Filter status
Search by stack name
Active
View nested
Stacks
NESTED
mastodon-yaoyi-a-HostedZone-PHO8JRVYXUCC
2026-03-22 00:59:05 UTC-0700
DELETE_FAILED
mastodon-yaoyi-a
2026-03-22 00:59:01 UTC-0700
DELETE_FAILED
c195402a5014886l13502043t1w22458096 9491
2026-01-16 16:15:11 UTC-0800
CREATE_COMPLETE
mastodon-yaoyi-a
Retry delete
Update stack
Stack actions
Stack info
Events
Resources
Outputs
Parameters
Template
Change sets
Git sync
Timeline view
The resource HostedZone is in a DELETE_FAILED state
Latest deployment timeline - new
View root cause
This is a timeline view of your latest stack deployment. If you start a new deployment, this view will reset.
Search events
Resources
Time


aws
Search
[Option+S]
United States (N. Virginia)
Account ID: 2245-8096-9491
voclabs/user4713919=wang.yaoyi@northeastern.edu
CloudFormation > Stacks > mastodon-yaoyi-a
Stacks (1)
Filter status
Search by stack name
Active
View nested
1
Stacks
c195402a5014886l13502043t1w2245809 69491
2026-01-16 16:15:11 UTC-0800
CREATE_COMPLETE
The resource WebService is in a CREATE_FAILED state
Latest deployment timeline - new
View root cause
This is a timeline view of your latest stack deployment. If you start a new deployment, this view will reset.
Search events
Resources
Time
Mar 22 02:04:00 -0700
Mar 22 02:05:00 -0700
Mar 22 02:06:00 -0700
Mar 22 02:07:00 -0700
Mar 22 02:08:00 -0700
Mar 22 02:09:00 -0700
Mar 22 02:10:00 -0700
Mar 22 02:11:00 -0700
Mar 22 02:12:00 -0700
Mar 22 02:13:00 -0700
Mar 22 02:14:00 -0700
Mar 22 02:15:00 -0700
WebService
StreamingApiTarget
Target
HttpListener
Database
Cache
Alb
ClientSg
Secret
Key
Vpc
AlbAccessLogBucket
Cluster
Alerting
CloudShell
Feedback
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